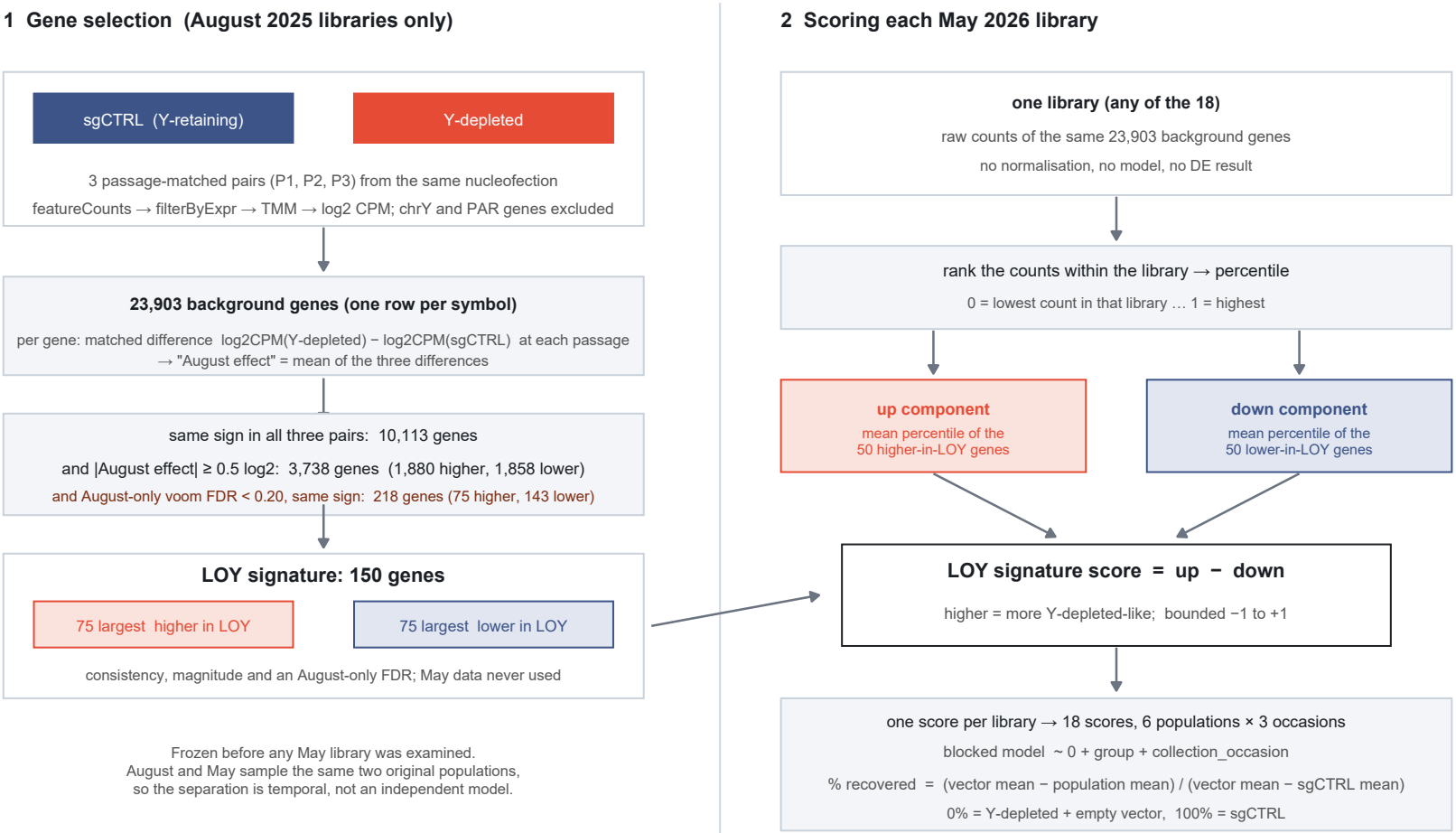

1 Gene selection (August 2025 libraries only)
sgCTRL (Y-retaining)
Y-depleted
3 passage-matched pairs (P1, P2, P3) from the same nucleofection
featureCounts → filterByExpr → TMM → log2 CPM; chrY and PAR genes excluded
23,903 background genes (one row per symbol)
per gene: matched difference log2CPM(Y-depleted) − log2CPM(sgCTRL) at each passage
→ "August effect" = mean of the three differences
same sign in all three pairs: 10,113 genes
and |August effect| ≥ 0.5 log2: 3,738 genes (1,880 higher, 1,858 lower)
and August-only voom FDR < 0.20, same sign: 218 genes (75 higher, 143 lower)
LOY signature: 150 genes
75 largest higher in LOY
75 largest lower in LOY
consistency, magnitude and an August-only FDR; May data never used
Frozen before any May library was examined.
August and May sample the same two original populations,
so the separation is temporal, not an independent model.
2 Scoring each May 2026 library
one library (any of the 18)
raw counts of the same 23,903 background genes
no normalisation, no model, no DE result
rank the counts within the library → percentile
0 = lowest count in that library … 1 = highest
up component
mean percentile of the
50 higher-in-LOY genes
down component
mean percentile of the
50 lower-in-LOY genes
LOY signature score = up − down
higher = more Y-depleted-like; bounded −1 to +1
one score per library → 18 scores, 6 populations × 3 occasions
blocked model ~ 0 + group + collection_occasion
% recovered = (vector mean − population mean) / (vector mean − sgCTRL mean)
0% = Y-depleted + empty vector, 100% = sgCTRL